PRT583 PROCESS DEVELOPMENT METHODOLOGIES

ASSIGNMENT 1

PROJECT PLAN

GROUP ALPHA

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Project Plan

1.Product vision

a. Product vision box

Name: Green Miles

Figure 1: Green Miles Concept Illustration - Before vs After Sustainable Transport Impact

Key Selling Points:	**Operating Requirements:**
• Tracks low-carbon transport such as walking, cycling, e-bikes, and e-scooters (also includes cars).	• Mobile platform. • GPS-based trip tracking and distance recording.

• Calculates and visualizes personal carbon emissions savings. • Provides incentives through carbon credits and community challenges. • Encourages sustainable travel behaviour through social engagement and rewards.	• Carbon emission calculation model for estimating savings. • Activity tracking system.

b. Elevator Test Statement

For environmentally conscious commuters, university students, and city residents who regularly choose low-carbon transport such as walking, cycling, or e-scooters instead of driving, and who want to understand and reduce their carbon footprint, Green Miles platform is a mobile transport tracking and carbon accounting application that automatically records sustainable travel using GPS and activity recognition, calculates carbon emission savings, and rewards users with verified carbon offset credits. Unlike traditional fitness applications such as Strava or Apple Fitness that primarily track exercise performance and physical activity, Green Miles focuses on measuring the environmental impact of daily transport choices and converting that impact into meaningful carbon savings. Our product transforms verified transport data into transparent, potentially tradeable carbon offsets through a digital marketplace, creating both environmental awareness and financial incentives for users while encouraging more sustainable transport behaviour.

2. Project objectives and its constraints

a. Project data sheet

i. **Clients/customers:**

1. Primary users: Daily commuters (students, employees), eco-conscious travellers.
2. Secondary customers: Organizations promoting sustainability (universities, NGOs, companies).

ii. Project leader:

Troy Kippen

iii. **Product manager:**

Troy Kippen

iv. **Executive Sponsors:**

Troy Kippen

v. **Project objective statement (POS):**

Develop Android carbon tracking app with GPS monitoring, emission calculation, and gamification features within 12 weeks

vi. **Business Objectives:**

- Increase sustainability awareness by showing emissions per trip and total impact.
- Drive behaviour changes via credits and leaderboard gamification.
- Create a rewards ecosystem (credits redeemable in in-app mart).
- Market Entry: Establish presence in $2.8B sustainability app market
- Social Impact: Facilitate 1 million kg CO2 equivalent reduction annually

vii. **Tradeoff matrix:**

Constraint	**Priority**	**Notes**
Scope	Medium (Adjustable)	MVP feature set can be trimmed (e.g., mart/leaderboard simplified) to meet deadline.
Schedule	High (Fixed)	Must be ready by submission/demo date.

Cost	High (Fixed)	Limited tools/cloud spend; use free tiers where possible.

viii. **Exploration factor:**

- Exploration factor: 6/10
  Reason: GPS/background tracking reliability, preventing cheating, emissions factor accuracy, credit economy tuning introduce uncertainty, user adoption uncertain

ix. Delay cost: The daily, weekly, or monthly cost of project delay

x. **Capabilities:** A list of the key capabilities (or features)

a. User Registration/Login/Logout
b. Choose transport mode (Car/Bus/Bike/Train/Walk)
c. User profile (name, totals: distance, emissions, credits)
d. Start/Stop trip tracking (GPS-based)
e. Real-time location monitoring and distance calculation
f. Algorithm to calculate Distance vs. Mode = Carbon Saved + Credits Earned.
g. Credit redemption for products/rewards
h. Community ranking and social comparison

xi. **Quality Objectives:** The project is expected to be delivered with a usable and stable state by the end of the 12-week timeline. The carbon calculation Accuracy should be reasonably close to accepted emission factors. The app should feel responsive during normal use enabling users to track trips without confusion. At the same time, the system should not crash under small to moderate usage, especially during GPS tracking sessions.

xii. **Performance/quality attributes:** The key performance and quality attributes focus on keeping the system usable, stable, and reasonably accurate during everyday use.

1. The system should respond quickly when users start or stop a trip, without noticeable lag.
2. GPS tracking should be stable and not drain too much battery during continuous use.
3. Trip distance and carbon calculation should stay reasonably accurate and consistent.
4. The application should remain stable during tracking and avoid crashes, even with network interruptions.
5. User data, especially location information, should be stored securely with proper authentication in place.
6. The interface should remain simple and easy to use without confusion.

xiii. **Architectural Guidelines:** From a design point of view, the system will follow a simple client–server structure. The mobile app handles tracking, while the backend processes trips and calculates carbon savings.

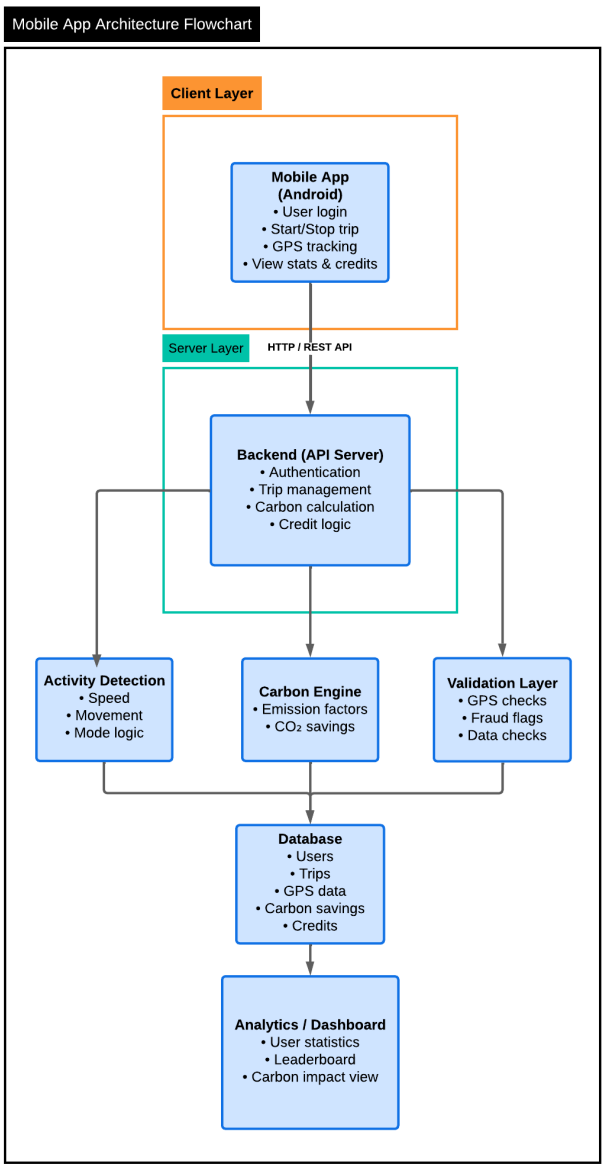

Figure 2: Mobile App Architecture Flowchart

xiv. Major project milestones:

1. Development of user authentication and profile features
2. Implementation of GPS-based trip tracking
3. Integration of carbon emission calculation logic
4. Development of credit/reward system
5. Basic UI for dashboard and statistics
6. Testing phase with sample data and bug fixing
7. Final system integration and demo preparation
8. Submission and final presentation

xv. **Issues/risks:**

Factors that could adversely impact this project are the following:

1. GPS tracking may produce inaccurate distance or location data
2. Users could attempt to fake trips to gain credits
3. Activity detection may not always correctly identify transport mode
4. Performance issues if too much data is processed at once
5. Privacy concerns related to storing user location data

3. Stakeholder Management

- **Stakeholders, Their Roles and Degree of Impact**

Stakeholder	**Category**	**Role in the Project**	**Impact Level**
Project Developers (Rahil, Meet, Musrat, Rafsan, Tran)	Internal	Designs, builds, tests, and maintains the application. Responsible for all technical decisions like GPS logic and database management.	Critical
Client (Troy)	Internal	Provides project requirements, project scope, approves deliverables and guides developers.	Critical
Academic Supervisor (Charles)	Internal	Provides academic guidance and evaluates all deliverables.	Critical
Users (Students, Working Professionals, etc.)	External	Register, log daily trips via GPS, earn carbon points, and redeem rewards through the app.	Critical
Environmental Organisations	External	Validate CO_2 saving calculations and provide domain knowledge.	Essential

Stakeholder	Category	Role in the Project	Impact Level
Transport Service Providers	External	Operators of buses, cycling, and e-scooter services whose transport modes are tracked and compared in the app.	Non-Essential
Local Government	External	Potential future adopter for public sustainability programs.	Non-Essential

- **Communication Channels and Frequency of Interaction**

Stakeholder	Communication Channel	Frequency
Developer	Emails, Teams Meetings, In person meetings	Twice a Week
Client	Email, Teams meetings	Weekly
Supervisors	Emails, In person Meetings	Weekly
Users	User Feedback through application	During testing
Environmental Organisations	Email, Project proposals	Frequently as needed
Local Governments	Email	As needed

4. Approach for delivery

Deliver the product:

The team will deliver the product through an iterative development process using Agile principles. The project will be divided into short development cycles called sprints, where specific components of the platform are designed, developed, and tested step by step.

In the early stages, the team will focus on building the core system architecture, user accounts, and GPS-based transport tracking. Later stages will develop additional features such as carbon emission calculations, activity recognition, social interaction features, and the carbon credit marketplace.

At the end of each sprint, the team will present a working prototype or completed feature to the client. This allows the client to review progress, provide feedback, and request improvements before the next sprint begins. Continuous testing, integration, and quality assurance will ensure that the system remains reliable and secure throughout development.

After all core features are completed and tested, the final product will be deployed as a mobile and web-based platform, followed by final review and delivery to the client.

The approach

The team will use an Agile Scrum approach to deliver the product. This approach allows the development team to build the platform in incremental stages while maintaining regular communication with the client.

The work will be organised into one-week sprints, where tasks such as design, development, and testing are completed for specific system features. After each sprint, the team will demonstrate the progress to the client and collect feedback to guide the next stage of development.

This approach ensures the project remains flexible, transparent, and responsive to client needs, while enabling the team to deliver a high-quality product that meets the project objectives.